

Ê PHASOR TEMPLATES

Wavenumber per medium *nonmag.*

k_i = (omega*sqrt(epsilon_r,i)/c) (rad/m) => k_2 = k_1*sqrt(epsilon_r,2/epsilon_r,1)

Direction table u in {x, y, z} = axis perp. to boundary

Table with 4 columns: Inc dir, Inc, Refl, Trans. Rows show +u and -u directions with corresponding e and e*jk terms.

+u direction <=> Med 2 at u >= 0. Reflected sign flips, transmitted matches incident.

General template (incident has pol. e-hat, amplitude a_0):

E-hat^i = a_0 e-hat e^+/-jk_1 u

E-hat^r = Gamma a_0 e-hat e^+/-jk_1 u

E-hat^t = tau a_0 e-hat e^+/-jk_2 u

(lossy med 2: replace e^+/-jk_2 u with e^+/-alpha_2 u e^+/-j beta_2 u)

Total in medium 1: E-hat_1 = E-hat^i + E-hat^r

e-hat by polarization plug into templates above

Table with 3 columns: Pol., e-hat, Note. Rows include Lin. x, Lin. y, Lin. at 45, RHC, LHC, and General cases.

Gamma, tau act on e-hat unchanged. Only swap the polarization, not the formulas.

POWER REFLECTION & TRANSMISSION

% reflected always

%Pr = 100 |Gamma|^2

% transmitted eta-ratio matters!

%Pt = 100 |tau|^2 * (eta_1/eta_2)

Check: %Pr + %Pt = 100.

If medium 2 is denser (eta_2 < eta_1), |tau| > 1 but %Pt < 100% because of the eta_1/eta_2 factor.

LOSSY MEDIA — 5 CASES

Loss tangent epsilon' = epsilon_r epsilon_0, epsilon_0 approx 1/(36pi x 10^9)

epsilon''/epsilon' = sigma/(omega epsilon_r epsilon_0)

Table with 4 columns: Type, sigma/omega epsilon, eta_c, alpha, beta. Rows include Lossless, Low-loss, Quasi-cond., Good cond., and Magnetic cases.

For Gamma/tau: drop j correction; use eta_0/sqrt(epsilon_r). Magnetic: keep mu_r.

Exact (quasi-cond) X = sqrt(1 + (epsilon''/epsilon')^2)

alpha = omega*sqrt(mu_epsilon'/2)*sqrt(X-1)

beta = omega*sqrt(mu_epsilon'/2)*sqrt(X+1)

theta_eta = 1/2 arctan(-sigma/(omega epsilon'))

BOUNDARY

eta_1, eta_2 — imp. (Omega)
Gamma — refl. coeff (unitless)
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k+normal
perp: E perp this plane
parallel: E parallel this plane

SNELL / OBLIQUE

theta_i, theta_t — angles from normal
n_i = sqrt(epsilon_r,i) — index of refr.
t_i — slab thickness (m)
d — lateral disp. (m)
plane of inc. = plane of k+normal
perp: E perp this plane
parallel: E parallel this plane

SNELL'S LAW — OBLIQUE INCIDENCE

Single boundary angles from normal

sin theta_2 = sin theta_1 * sqrt(epsilon_r1/epsilon_r2) = sin theta_1 * (n_1/n_2)

Multilayer chain stack of slabs 1 -> 2 -> 3 -> 4

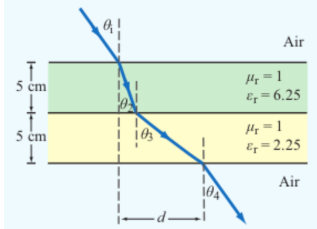
sin theta_{i+1} = sin theta_i * sqrt(epsilon_{r,i}/epsilon_{r,i+1})

Air-slabs-air shortcut outer media identical

theta_exit = theta_incident

Snell only sets angles. Reflection/transmission magnitudes need Fresnel coefficients (parallel vs perpendicular).

LATERAL DISPLACEMENT



Beam offset through N slabs thickness t_i, refracted angle inside slab i

Slab-indexed theta_i = angle inside slab i

d = sum_{i=1}^N t_i tan theta_i

Angle-indexed (HW3 P3 style) theta_1 = incident, theta_{i+1} = inside slab i

d = sum_{i=1}^N t_i tan theta_{i+1}

Same physics, different numbering. In HW3 P3: slab 1 contributes t_1 tan theta_2, slab 2 contributes t_2 tan theta_3. The angle in the formula is always the one the ray makes inside that slab.

Air-slab-air: exit angle returns to theta_1 but is laterally shifted by d.

TABLE 8-2 — Gamma, tau, R, T SUMMARY

Med. 1 (z < 0) incident side; Med. 2 (z >= 0) transmitted. Shorthand: c_i = cos theta_i, c_t = cos theta_t.

Table with 3 columns: Normal (theta_i = 0), perp pol, parallel pol. Rows include Gamma, tau, R, T, and R+T for both media.

Notes: sin theta_t = sqrt(mu_1 epsilon_1 / mu_2 epsilon_2) sin theta_i; nonmag. => eta_2/eta_1 = n_1/n_2.

Intrinsic impedance eta_i plug into table above

eta_i = sqrt(mu_r/epsilon_r) eta_0 * (mu_r=1) -> eta_0/epsilon_r,i (Omega) eta_0 = 120pi approx 377 Omega

Default mu_r = 1 (air, dielectric). Use mu_r != 1 only if problem says "ferromagnetic"/iron/ferrite/etc.

Low-loss correction eta_c sigma/(omega epsilon) <= 1

eta_c approx sqrt(mu_epsilon'/epsilon') * (1 + j sigma/(2 omega epsilon)) -> drop j -> eta_0/sqrt(epsilon_r)

For Gamma/tau just use eta_0/sqrt(epsilon_r). The j term is for |eta_c|, theta_eta. See LOSSY MEDIA — 5 CASES for all regimes.

RECTANGULAR WAVEGUIDE — MODES

TE mode E_z = 0, H_z != 0

TM mode H_z = 0, E_z != 0

Dimensions a x b with a > b along x-hat, y-hat; wave in z-hat.

E_x form (TE) read m, n from arguments

E_x proportional to cos((m*pi*x)/a) sin((n*pi*y)/b) sin(omega*t - beta*z)

coef. on x = m*pi/a; coef. on y = n*pi/b.

Identify TE_{mn} vs TM_{mn} cos/sin pattern of E_x is identical for both!

1. Read the problem statement (e.g. "a TE wave" -> TE). 2. If m=0 or n=0 in the mode label => must be TE (TM forbids 0).

3. Source field H_z(x, y) given => TE. Source field E_z(x, y) given => TM.

TE allows at least one of m, n >= 1 (other can be 0). TM needs both m, n >= 1. That's why TE_{10} exists but TM_{10} doesn't.

TABLE 8-3 — FULL FIELD COMPONENTS

Shorthand: C_X = cos(m*pi*x/a), S_X = sin(m*pi*x/a), C_Y = cos(n*pi*y/b), S_Y = sin(n*pi*y/b). All fields carry an e^{-j beta z} factor (omitted below). k_c^2 = (m*pi/a)^2 + (n*pi/b)^2.

TE mode generator: H-hat_z

H-hat_z = H_0 C_X C_Y

E-hat_x = (j omega mu / k_c^2) * (n*pi/b) H_0 C_X S_Y

E-hat_y = - (j omega mu / k_c^2) * (m*pi/a) H_0 S_X C_Y

E-hat_z = 0, H-hat_x = -E-hat_y/Z_{TE}, H-hat_y = E-hat_x/Z_{TE}

TM mode generator: E-hat_z

E-hat_z = E_0 S_X S_Y

E-hat_x = - (j beta / k_c^2) * (m*pi/a) E_0 C_X S_Y

E-hat_y = - (j beta / k_c^2) * (n*pi/b) E_0 S_X C_Y

H-hat_z = 0, H-hat_x = -E-hat_y/Z_{TM}, H-hat_y = E-hat_x/Z_{TM}

TEM (plane wave) for reference: E-hat_x = E_0 e^{-j beta z}, H-hat_y = E-hat_x/eta, f_c N/A, k = omega/sqrt(mu_epsilon).

CUTOFF FREQUENCY

f_{mn} m, n integers >= 0 (not both zero)

f_{mn} = u_{p0}/2 * sqrt((m/a)^2 + (n/b)^2) (Hz)

Common cutoffs (a > b)

Table with 2 columns: Quantity, Formula. Rows include u_{p0}, TE_{10}, TE_{01}, TE_{20}, and TE_{11}, TM_{11} cutoff frequencies.

Propagation requires f > f_{mn}.

epsilon_r FROM DISPERSION

Given omega, beta, mode (m, n)

epsilon_r = (c^2/omega^2) * [beta^2 + ((m*pi/a)^2 + (n*pi/b)^2)]

Sanity: beta in rad/m, a, b in m, omega in rad/s.

WAVE IMPEDANCE & H_y

Z_{TE} = eta / sqrt(1 - (f_c/f)^2) (Omega) Z_{TM} = eta sqrt(1 - (f_c/f)^2)

eta = eta_0/epsilon_r H_y = E_x/Z_{TE} (TE mode)

Z_{TE} > eta always (denominator < 1); Z_{TM} < eta.

GROUP VELOCITY & TRAVEL TIME

u_g = u_{p0} sqrt(1 - (f_{mn}/f)^2) (m/s)
u_p = u_{p0} / sqrt(1 - (f_{mn}/f)^2) (m/s) => u_p u_g = u_{p0}^2 (m^2/s^2)
t = L/u_g (s) travel time through guide of length L
Higher modes have lower u_g => they arrive later. Group delay spread = sum over excited modes.

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WAVEGUIDE

a, b — wide, narrow dim. (m);
a > b
m, n — mode indices (int.)
f_{mn} — cutoff (Hz)
mu — prop. const (rad/m)
Z_{TE}/Z_{TM} — wave imp. (Omega)
E_x, H_y — field comps (V/m, A/m)
eta = eta_0/sqrt(epsilon_r)

VELOCITIES

u_{p0} = c/sqrt(epsilon_r) — bulk phase vel.
u_p = u_{p0} / sqrt(1 - (f_c/f)^2)
u_g = u_{p0} sqrt(1 - (f_c/f)^2)
u_p u_g = u_{p0}^2
t = L/u_g — travel time
c = 3 x 10^8 m/s

POWER & UNITS

%Pr = 100 |Gamma|^2
%Pt = 100 |tau|^2 eta_1/eta_2
%Pr + %Pt = 100
sigma (S/m), epsilon (F/m), mu (H/m)
epsilon_0 = 8.85 x 10^-12 F/m
mu_0 = 4pi x 10^-7 H/m
1/(36pi) x 10^-9 approx epsilon_0